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Enhancement of CF/PA66 ultrasonic welding joints via transforming 1D to 2D profiled in-situ laser-etched energy director structure



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Highlights

- This study employed in situ laser ablation to rapidly fabricate diverse grooved ED structures on CFRTP plates.
- The rapid heating and cooling of laser ablation modifies the ED crystallinity, aids deformation, and boosts heat production.

- A finite element model was used to analyze the temperature and stress distributions in joints with various ED structures.
- The strengthening mechanism of laser etched ED was discussed in detail.

Abstract

Ultrasonic welding has garnered increasing attention in the aerospace and automotive sectors as an efficient and environmentally friendly joining technology for carbon fiber reinforced thermoplastic composites (CFRTP). The design of energy directors (EDs) is crucial, as they can effectively concentrate interfacial heat generation and stabilize the welding process, thereby enhancing the welding quality. In this study, in-situ laser etching technology was employed to rapidly fabricate diverse groove-typed ED structures on the surface of CF/PA66 sheets, stepwise improving the mechanical properties of welded joints by changing energy director structure. The strengthening effect was verified through interfacial morphology characterization, mechanical property testing, and fracture analyses. Furthermore, finite element thermomechanical coupling modelling was utilized to analyze the temperature/stress distribution and transformation during the welding processes for the joint without ED, with 1 direction (1D) groove type ED and 2 direction (2D) grid ED structures, further clarifying their strengthening effects. The results indicate that the laser-etched energy director structures concentrate the welding energy, significantly reducing the dispersion of the weld seam distribution and increasing the proportion of the central weld seam, thereby remarkably enhancing the joint failure load. Specifically, the joint strengthening effect exhibited a gradual improvement trend from groove EDs to grid EDs, with the joint strength increasing by 35.2% and 64.4%, respectively, compared to that of joints without ED. The interface strengthening mechanism of the 1D to 2D ED structures mainly involves three core aspects: (i) Enhancement of heat generation caused by the change of crystallinity from laser etched 1D ED structure to 2D ED structure (ii) Better Concentration of welding heat and improvement of thermal uniformity caused by 2D grid ED structure. (iii) Better interface bonding and edge defects eliminating attributed to the increasement of material flowing channel caused by improved 2D ED structure.



Keywords

CFRTP; Ultrasonic welding; In-situ laser etched; Energy director structure; Strengthening mechanism

1. Introduction

With the growing global demand for a green economy, the manufacturing industry has identified energy conservation and emission reduction as critical objectives [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#). Currently, the development of lightweight structure is regarded as one of the most effective approaches to achieve this. Carbon fiber reinforced thermoplastic (CFRTP) composites have emerged as next generation lightweight materials owing to their exceptional mechanical properties, weldability, reparability, and recyclability [\[6\]](#), [\[7\]](#), [\[8\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#). As CFRTP finds increasingly widespread applications in the aerospace, automotive, and railway transportation sectors, its joining technology has become crucial [\[9\]](#), [\[12\]](#), [\[13\]](#), [\[14\]](#), [\[15\]](#). Among the various joining techniques for CFRTP components, welding processes have attracted significant attention owing to its high automation level, low cost, and high efficiency. Common CFRTP welding methods include resistance, induction, ultrasonic, and laser welding [\[16\]](#), [\[17\]](#). Ultrasonic welding is recognized as one of the most ideal CFRTP joining technologies owing to its energy efficiency, environmental friendliness, and cleanliness [\[18\]](#), [\[19\]](#), [\[20\]](#), [\[21\]](#). Nevertheless, conventional ultrasonic welded joints suffer from insufficient strength, therefore requires the prefabrication of small protruding structures (referred to as energy directors, EDs) on the material surface [\[22\]](#), [\[23\]](#). These ED structures can effectively concentrate ultrasonic energy to enhance joint quality.

However, fabricating conventional energy directors requires reserved mold positions during mold opening, a cumbersome process that increases production costs while compromising manufacturing efficiency [\[24\]](#), [\[25\]](#), [\[26\]](#), [\[27\]](#), [\[28\]](#). To address the inherent limitations of traditional EDs, researchers have proposed various innovative design approaches, highlighting the urgency of developing novel ED structures to advance CFRTP ultrasonic welding technology. Villegas et al. pioneered the ultrasonic welding of planar EDs, simplifying the fabrication process of ED structures [\[27\]](#). Palardy et al. systematically investigated the correlation between flat ED thickness and heat generation characteristics, thereby improving welding

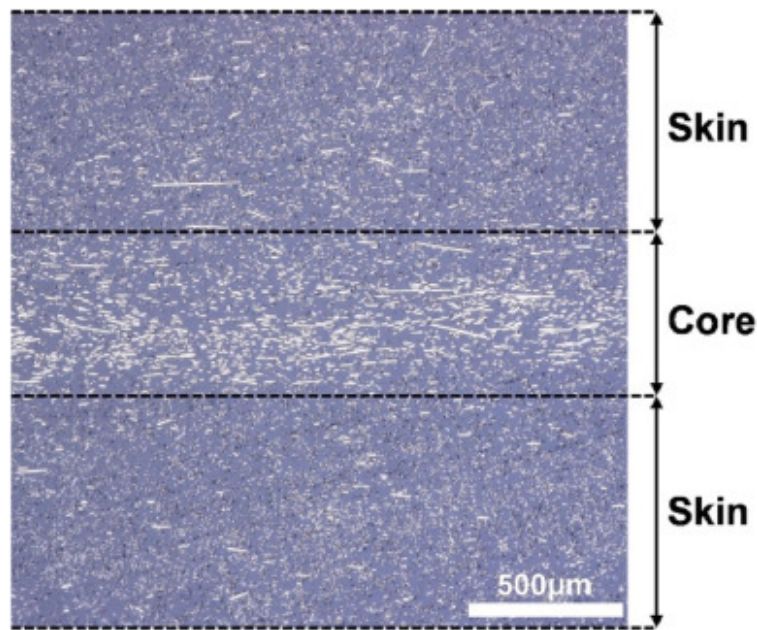
uniformity. However, this approach suffers from the difficulty in controlling the initiation site of molten pools [29]. Li et al. fabricated stainless steel grid EDs and studied the effect of pore size on the performance of CF/PA6 joints, enhancing interfacial heat generation [30]. Yet the heterogeneous ED tends to exacerbate pore formation. Wang et al. fabricated external embossed EDs with a protruding grid structure via ultrasonic embossing technology [31]. Although this configuration demonstrated a superior energy concentration capability, its fabrication requires a relatively large amount of resin material. These techniques all rely on external addition, in contrast, the direct surface structuring of welding sheets for in-situ ED fabrication may serve as an innovative strategy for ED design in ultrasonic welding. In addition to the design and optimization of energy director structures, the interfacial strengthening mechanisms induced by these structures are equally pivotal. Finite element simulations provide critical insights into the thermomechanical interactions during complex ultrasonic welding processes, enabling a systematic analysis of the energy dissipation and heat generation mechanisms [32]. Villegas and Levy's research teams compared the thermal field characteristics between triangular and planar EDs on the COMSOL platform, where the localized temperature elevation of the EDs during CF/PEI ultrasonic welding effectively demonstrated the interfacial energy concentration effect [22,33,34]. Wang et al. investigated the influence of cross sectional area and vertex angle of triangular EDs on the temperature distribution of CF/PEEK ultrasonic welding, revealing the sensitivity of heating rate to vertex angle variations [35]. However, current research primarily focuses on ultrasonic welding processes without ED, with insufficient studies on the interfacial energy concentration characteristics and thermomechanical coupling mechanisms of EDs in CFRTP ultrasonic welding.

Inspired by surface processed ED technologies, this study proposes an innovative in-situ ED fabrication method for the ultrasonic welding of CFRTP. In situ in line and grid shaped groove structures were fabricated on CFRTP surfaces via laser etching, systematically investigated the effects of laser-etched EDs on the mechanical properties of joints. The strengthening mechanism of these in-situ groove EDs was elucidated by constructing a three dimensional thermomechanical coupling finite element model and comparing the resulting interfacial temperature and stress fields. Comparative analysis revealed that laser etching reduced the interfacial crystallinity and promoted deformation. Furthermore, the groove geometry effectively concentrated heat and facilitated the mechanical interlocking of the material at the weld edges. This approach pioneers a novel process route for the ultrasonic welding of CFRTP.

2. Experimental procedures and modeling setup

2.1. Materials

The materials used in this experiment were 30wt% carbon fiber-reinforced polyamide 66 (CF/PA66) plates supplied by Hefei Jishijie Co., Ltd. The average diameter and length of the carbon fibres were approximately 7 and 200 μm , respectively. Owing to the varying fibre orientations induced by injection moulding, some carbon fibres could not be fully exhibited. Its microstructure along the lap length is shown in Fig. 1, where the white dots represent the carbon fibers. The carbon fibers exhibited a "skin-core-skin" structure: fibers in the "skin" regions are parallel to the injection direction, while those in the "core" region were perpendicular to it. This structure was induced by the injection moulding process and has also been reported in other studies related to injection molded plates [36,37]. The CF/PA66 sheets used in the experiments were fabricated by injection moulding and then cut into dimensions of 60 mm^L×30 mm^W×3 mm^H specimens to meet the requirements of the subsequent ultrasonic welding. The material properties of the CF/PA66 composite are detailed in Table 1, providing support for subsequent welding process optimization, joint mechanical performance analysis, and finite element simulation. Combining the high strength of carbon fibers with the excellent processability and environmental stability of the matrix, CF/PA66 offers significant advantages in the application of lightweight structural components.



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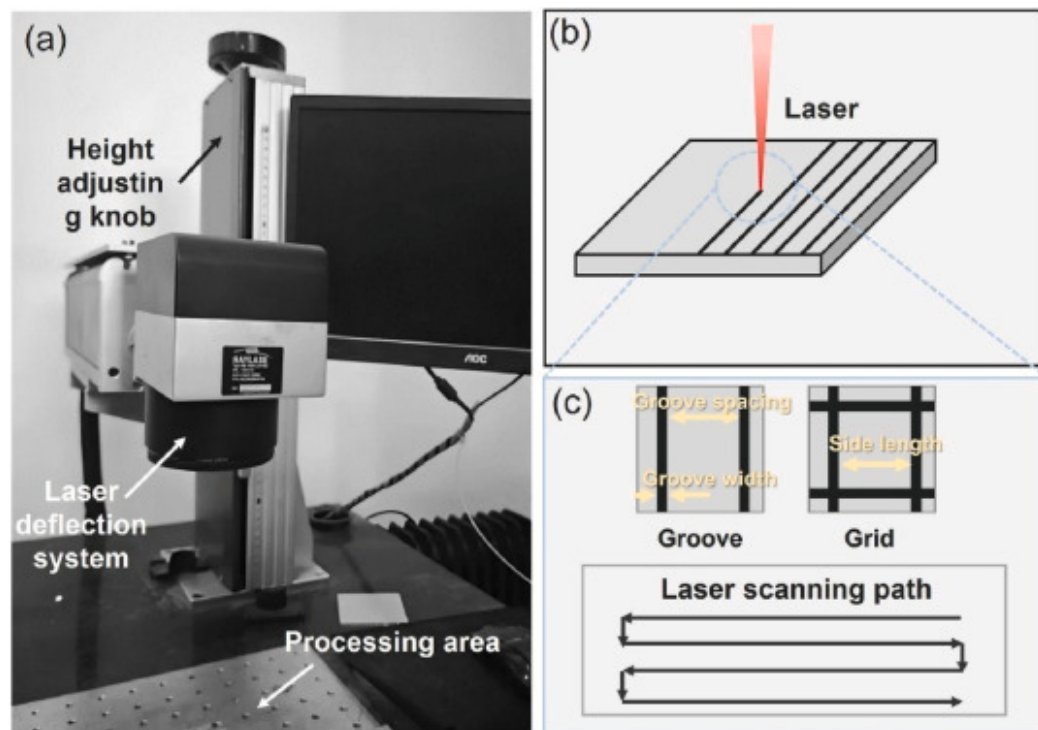
Fig. 1. Microstructure of the plates along the lap length of the plate.

Table 1. The material properties of CF/PA66 [38].

Melting point (°C)	Density (g/cm ³)	Glass transition temperature (°C)	Decomposition temperature (°C)	Tensile strength (MPa)
264	1.224	78.9	352.6	265

2.2. Laser etching process

In this study, laser etching technology was used to fabricate ED structures on CFRTP plates. The experimental setup utilized an SPI Lasers SP-070P-A-EP-Z-F-Y nanosecond pulse laser system (UK), featuring a 7 ps ultra short pulse duration and 1064nm near infrared wavelength, enabling micron level precision in surface patterning. As illustrated in Fig. 2 (b), the laser focus position was regulated using precision optical systems to translate predefined geometric patterns into controlled scanning paths (Fig. 2 (c)). Under the optimized laser parameters (detailed in Table 2), the laser beam achieved precise material ablation, selectively removing a certain amount of material to form groove structures. The etched surface features regular groove structures, the raised portions of which serve directly as EDs during ultrasonic welding. Two ED configurations were fabricated: parallel grooves (1D) and grid patterns (2D), both with a spacing (groove spacing/grid side length) of 1 mm and a groove width of 0.2mm.



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Fig. 2. ED process and equipment: (a) laser generation system, (b) laser etching process, (c) ED structure laser etching design and laser scanning path.

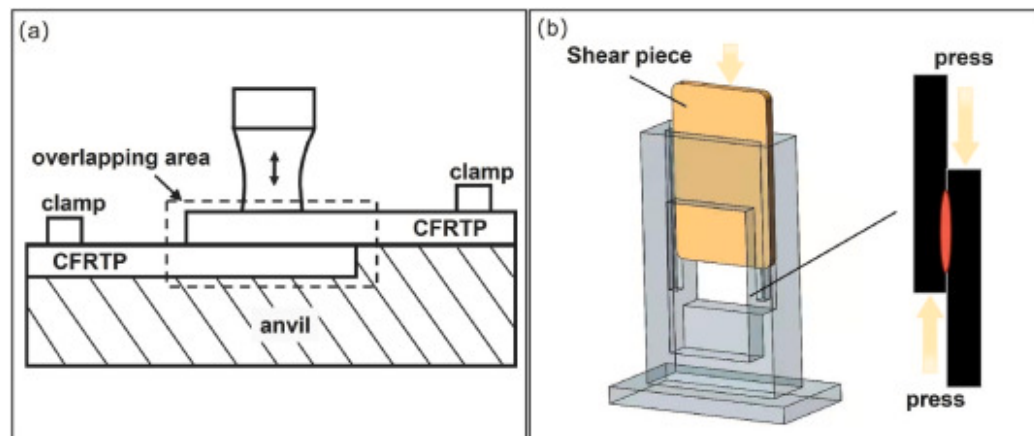
Table 2. Laser etching parameters for ED processing.

Laser process parameters	Unit	Value
Laser power	W	70
Scanning speed	mm·s ⁻¹	2000
Pulse duration	ns	10

Laser process parameters	Unit	Value
Pulse frequency	kHz	25
Single pulse energy	mJ	0.7
Spot diameter	μm	50

2.3. Ultrasonic welding process

Ultrasonic welding experiments were conducted using a 20kHz high frequency ultrasonic spot welder with a nominal power of 3600W. Characterized by a stable power output and precise vibration frequency, the equipment provides sufficient and controllable ultrasonic energy for the welding of CF/PA66 composites and is equipped with a cylindrical horn with a diameter of 20mm. All the welding specimens were joined in a lap shear configuration with an overlapping area of 30mm×30mm. During welding, the sheet with the ED structures was positioned as the lower plate, as illustrated in Fig. 3(a). This arrangement facilitates the formation of an interlocking structure between the flowing molten material of the upper plate and the lower plate. Given the critical influence of vibration time on ultrasonic welding, a time controlled welding mode was adopted with the following parameters: welding time, 1 s, amplitude, 35μm, and pressure, 0.2MPa [39]. Three distinct welding configurations were designed and tested: no ED specimens, linear groove ED specimens (1D), and grid ED specimens (2D). Multiple replicate specimens were prepared under each parameter condition: three for shear strength testing and the rest for microstructural characterization (including interfacial microtopography observation, fracture analysis, and crystallinity measurement) and temperature monitoring during welding using thermocouples.



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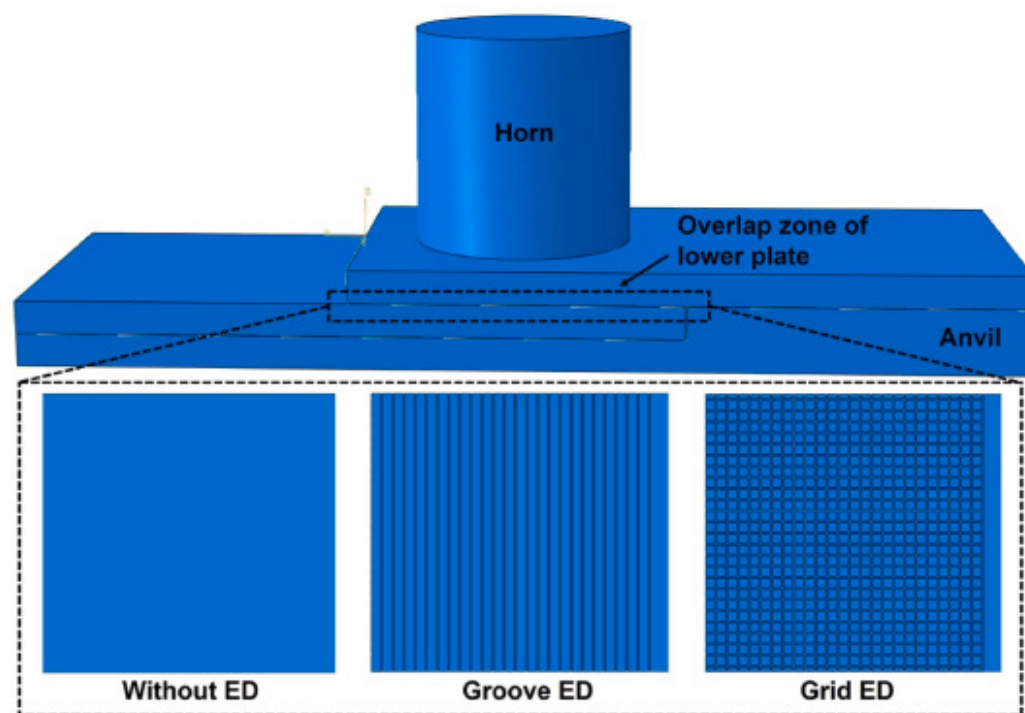
Fig. 3. Experimental set-up: (a) the welding set-up, (b) the test of mechanical property.

2.4. Testing and analysis

Surface 3D topography observation of the laser processed EDs was performed using an Olympus DSX510 optical microscope, which was also employed to examine the weld cross sectional and fracture surface morphologies. Compression shear tests on the welded joints were conducted using an Instron 5967-30KN universal testing machine at a loading rate of 1 mm/min, with the setup configuration illustrated in Fig. 3(b). Differential Scanning Calorimetry (DSC) was performed using a TA Instruments DSC25 to analyze the thermal properties of the matrix material, the material near the laser-ablated structure, and the material in the weld zone (The matrix material was cut into small pieces for testing, while a small amount of powder was scraped off from the region near the laser ablated structure and the weld zone using a scraper for testing). The temperature variation at the center of the welding interface was monitored using a distributed temperature measurement system consisting of a K-type thermocouple (Omega TJ36-CAXL-116U-12) and an NI-9213 high-precision data acquisition module. Before testing, a through-hole was pre-drilled in the lower plate, the thermocouple wire was passed through the hole, and the sensor was fixed to the testing interface.

2.5. Finite element model

A finite element model for the ultrasonic welding of CF/PA66 was developed using ABAQUS Explicit. As shown in Fig. 4, the model comprises three core components: an anvil that serves as the load-bearing base, a horn that delivers high frequency ultrasonic vibration and welding pressure, and two rectangular CF/PA66 composite plates to be joined. The element types were selected based on their functional roles and computational requirements. The CF/PA66 plates, which are critical regions for temperature evolution, viscoelastic deformation, and interfacial joining, were discretized using C3D8T elements to ensure precise thermal capture. The horn and anvil, which primarily function to transmit vibrations and support the load, were modeled as rigid bodies using R3D4 elements.



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Fig. 4. Ultrasonic welding geometric assembly diagram.

Based on this framework, three finite element models with distinct welding configurations were constructed to systematically compare the influence of the energy director structures: (i) a flat interface model without an ED, corresponding to the

unstructured specimens in the experiment, (ii) a model featuring a single groove ED structure; and (iii) a model incorporating a grid ED. Numerical simulations and comparative analyses of these three models provide a direct visualization of the differences in temperature distribution and evolution among the various ED designs.

Through DSC testing and calculations, the CF/PA66 composite with 30wt% carbon fiber content exhibited a density of 1.224g/cm^3 and a thermal conductivity of $3.313\text{W/(m}\cdot\text{K)}$. The characterization of the viscoelastic behavior adopted the frequency data obtained from the DMA tests. The data were input into the material model, and the parameters of the Prony series model were derived through normalized calculations via the Abaqus material evaluation, as shown in Table 3, N is the number of Prony series. After normalization processes the data from stress. The finite element simulation process for CF/PA66 ultrasonic welding established in this study was divided into two steps. Step 1 was the initial contact phase, where the horn was simulated to move downward until it contacted the upper sheet surface and established a preset trigger force. Step 2 is the ultrasonic welding phase, after the stable trigger force is established, the horn applies a high frequency ultrasonic vibration at a frequency of 20kHz. The horn displacement in this phase was precisely defined using periodic displacement constraints to simulate the vibration characteristics of the horn during actual welding. To clearly capture the dynamic evolution of the interfacial temperature field during welding and accurately analyze the differences in heat generation efficiency and temperature distribution across different phases, the finite element model was set to output calculation results at an extremely small-time increment of 0.001 s. Using high density time node data, the model can fully reproduce the dynamic characteristics of the interfacial temperature, such as increase, diffusion, and stabilization, throughout the entire welding process from initiation to completion.

Table 3. Parameters of the generalized Prony model.

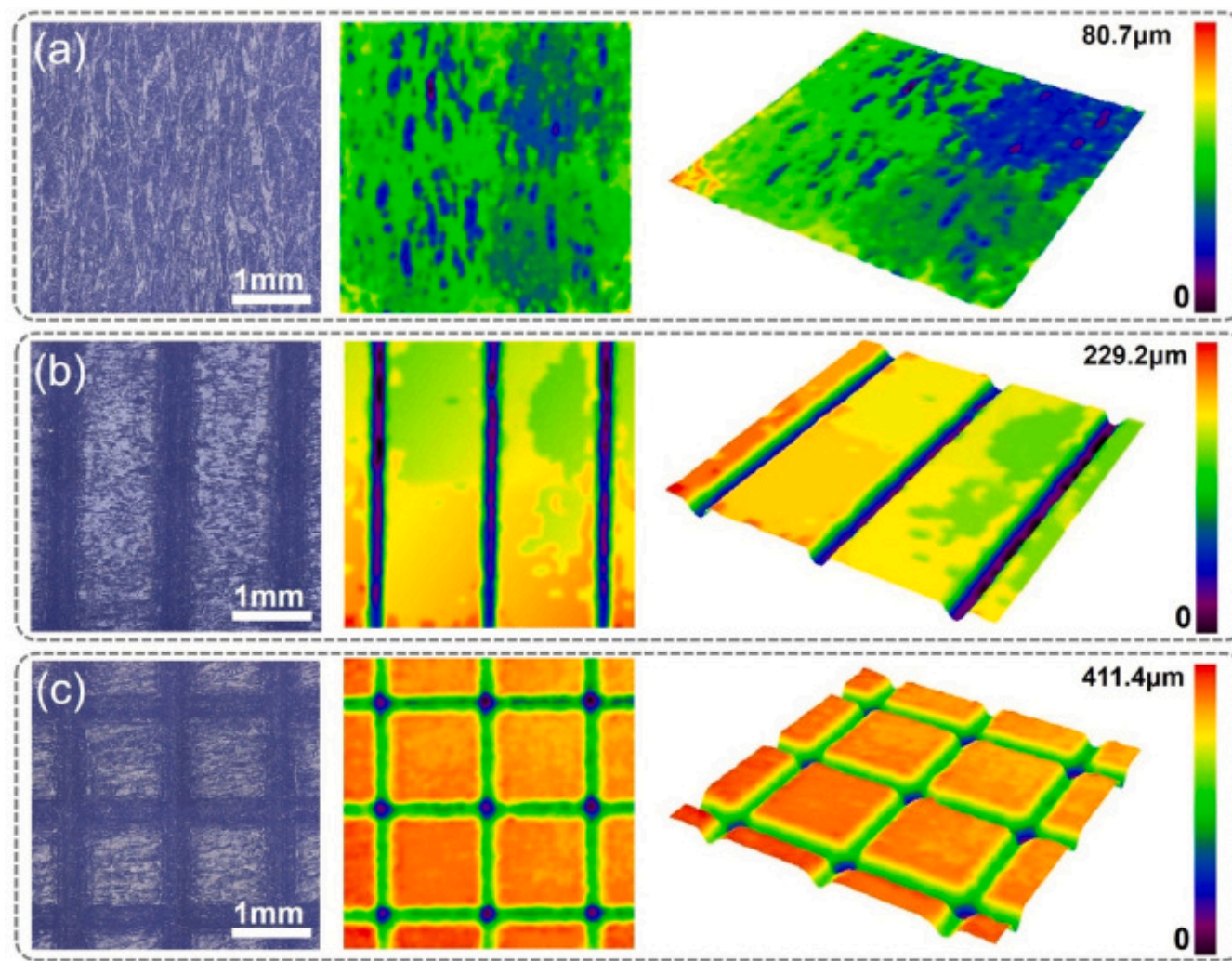
i	Relaxation modulus	Relaxation time
1	0.0716	0.00142
2	0.0648	0.0135
3	0.0591	0.0176
4	0.0624	0.0334
5	0.0592	0.214

i	Relaxation modulus	Relaxation time
6	0.0538	0.518
7	0.0334	13.508
8	0.2	317.6

3. Results and discussion

3.1. Change of surface morphology and crystallinity by laser-etched ED

[Fig. 5](#) presents the surface morphological characteristics of the sample without ED and two types of ED structures: linear groove (1D) and grid EDs (2D). After laser etching, the single groove depth of both linear groove and grid EDs was controlled at 0.2mm, aligning with the optimal ED thickness parameters reported [\[24\]](#), ensuring structural strength while providing adequate space for molten material flow and interfacial joining. Both types of EDs adopted a groove spacing of 1 mm to guarantee a reasonable structural arrangement. In comparison, grid EDs not only have more grooves than linear groove EDs but also form 0.4mm deep depressions at groove intersections owing to repeated laser etching.



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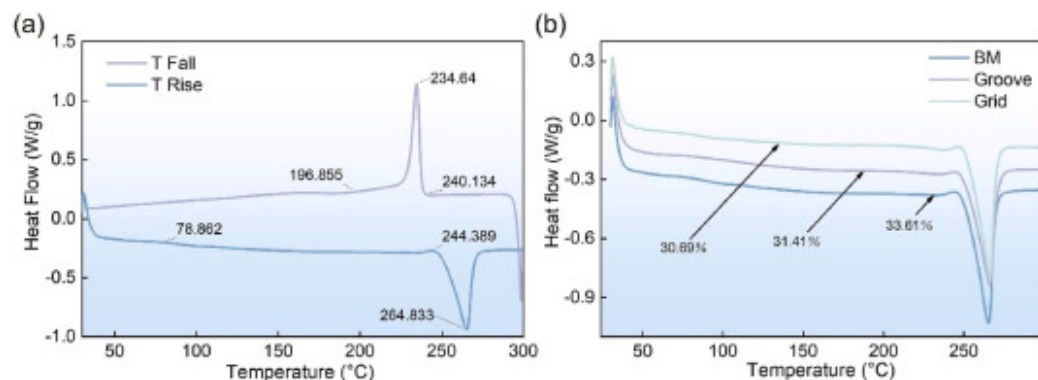
Fig. 5. surface morphology of different ED structures: (a) Original surface, (b) groove structure surface (1D), (c) grid structure surface (2D).

During in situ laser etching, a high energy laser beam acts instantaneously on the surface of the CF/PA66 composite, producing a significant thermal effect. The surface PA66 matrix is rapidly heated and vaporized, undergoing selective removal to form an

energy director structure. Simultaneously, some exposed carbon fibers inside the matrix were ablated, introducing oxygen containing polar functional groups and increasing the surface roughness of the fibers. The change in surface roughness provides favorable conditions for the formation of mechanical interlocking structures during the subsequent ultrasonic welding process [40,41]. Fig. 6 presents the thermal property analysis of the materials. Differential Scanning Calorimetry (DSC) was performed on the laser processed ED structured materials to obtain heat flow temperature curves for different ED configurations, as shown in Fig. 6(b). The degree of crystallinity (DoC) of the processed materials was calculated from these curves using the following formula [42]:

$$\text{Doc (\%)} = \frac{\Delta H_m}{(1-wf)\Delta H_m^0} \times 100 \quad (3-1)$$

where ΔH_m represents the melting enthalpy measured during the heating cycle (unit: J/g), wf denotes the mass fraction of short carbon fibers (30% carbon fiber content in this study, corresponding to 0.3), is the melting enthalpy of fully crystalline nylon 66 (reference value: 190J/g). The calculated DoC values for the without ED, groove type (1D), and grid type ED (2D) configurations were 33.61%, 31.41%, and 30.69%, respectively. As indicated in Fig. 6(a), during the cooling process of the CF/PA66 composite, a crystallization peak appeared when the temperature reached 240°C, indicating that the polymer began to crystallize below this temperature, underwent rapid crystallization at 234.6°C, and completed crystallization at 196.9°C. Polymer crystallization is essentially an ordering process of melted molecular chains, which requires a specific duration. The time interval between 240°C and 196.9°C is particularly critical for the crystallization of CF/PA66. Owing to the rapid heating and cooling characteristics, the material near the laser processed structures undergoes melting, followed by recrystallization. Meanwhile, the cooling rate of CF/PA66 within the crystallization temperature range reached 35.6°C/s, which was much higher than that of the injection molded CF/PA66 composite in the same temperature interval. This results in a decrease in the crystallinity of the material adjacent to the etched ED structure, and the reduced crystallinity lowers the stiffness at geometric changes, making it more prone to strain during the welding process, thereby enhancing interfacial heat generation, as shown in Equations (3)–(3) [43].



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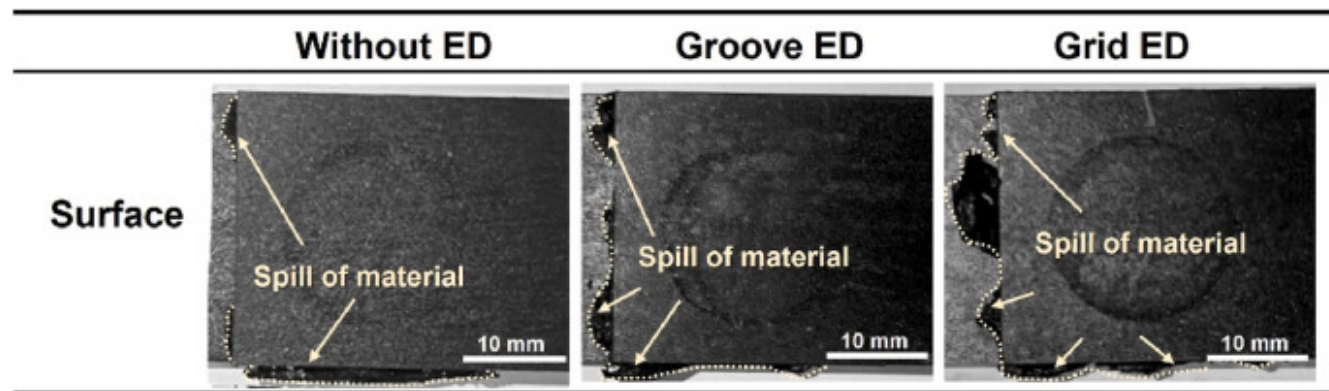
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Fig. 6. DSC test results: (a) Base Material, (b) without ED structure and different structure by laser.

3.2. Weld formation of different ED structure

Fig. 7 shows a direct comparison of the material overflow morphology of the three joints (without ED, linear grooved ED (1D), and grid ED (2D)), illustrating the effect of adding ED structures on the material flow behavior after welding. It can be observed that the surface indentations of the three joints are not obvious and show little difference. This is mainly attributed to the identical welding parameters used in the experiment and the unchanged contact conditions between the horn and the upper plate surface. Notably, considerable differences exist in the spilling of material at the edges (marked by yellow dashed lines) among different joints, which is related to the material flow behavior. Compared with the joint without ED, both the linear groove and grid ED joints exhibited more obvious material extrusion traces, with molten material squeezed out from the joint edges under welding pressure. This phenomenon directly confirms that ED structures enhance the welding energy concentration: ultrasonic energy scatters at the interface of the joint without ED, whereas ED structures, through the geometric guidance of grooves, concentrate ultrasonic energy, effectively reducing energy loss. This significantly intensified the interfacial material melting and increased the total amount of molten material. Under the constant pressure applied by the horn, excess molten material overflowed outside the joint. Notably, the material extrusion volume of the grid ED joint was significantly larger than that of the linear groove ED joint. This difference highlights the advantage of the grid structure during welding: compared

with unidirectional linear grooves, the cross design of the grid structure enables the molten material to flow uniformly and spread adequately along the grid grooves [44].



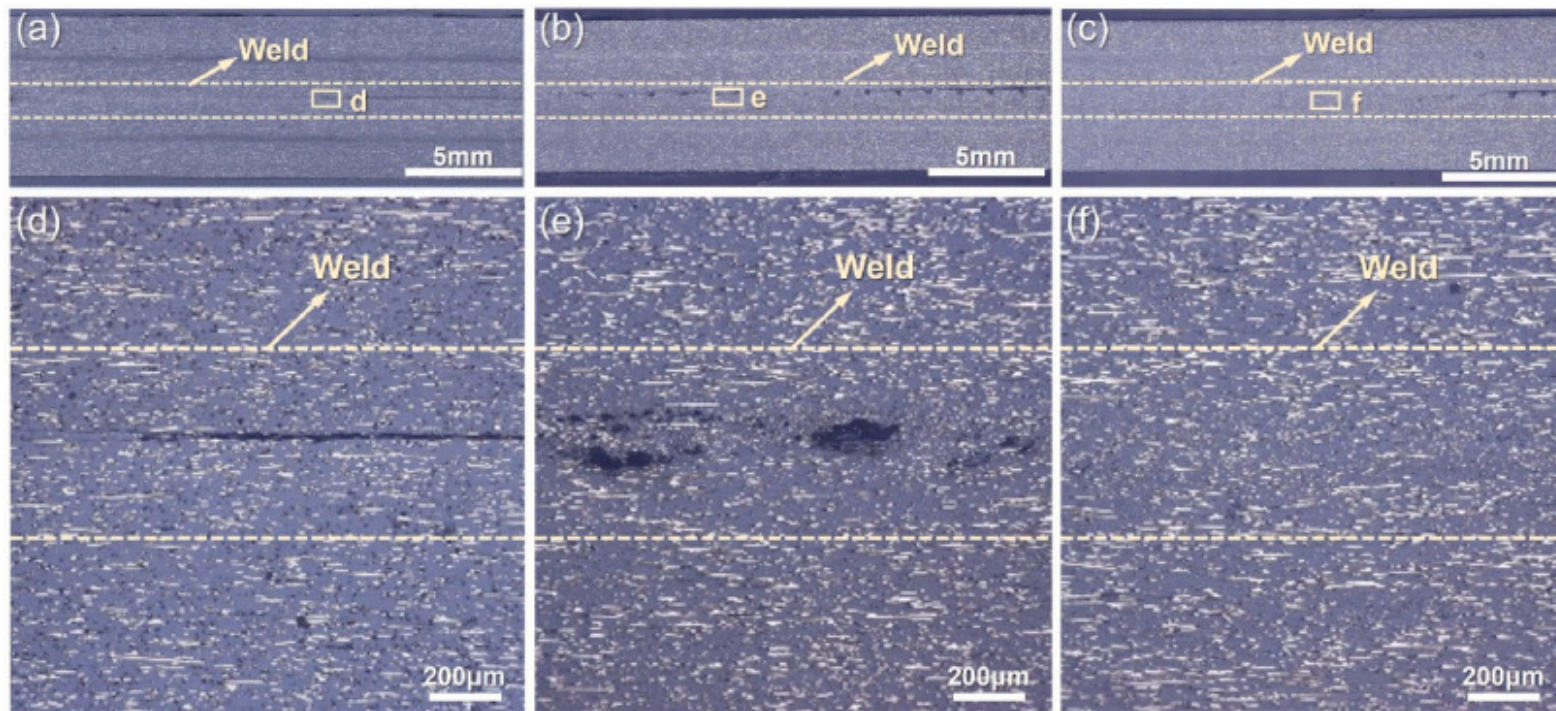
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Fig. 7. Material overflow morphology of joints with different EDs.

Fig. 8 shows the interfacial microstructures of the three joints: without ED, with linear groove ED (1D), and with grid ED (2D). The essence of interface joining in composite ultrasonic welding is a synergistic process of rapid melting, uniform mixing, and crystallization of surface materials, and its microstructural evolution directly determines joint quality. As shown in the figure, the joint without ED suffers from scattered ultrasonic energy and low interfacial utilization efficiency during welding. Under the preset short welding time, the molten material at the interface is insufficient to fill the interfacial gaps fully, resulting in a large number of unjoined regions that become potential weak points for subsequent mechanical properties. In contrast, joints with linear groove and grid EDs achieved a directional concentration of ultrasonic energy through the structural guidance of the EDs, leading to a significant increase in the effective interfacial joined areas and demonstrating superior energy focusing effects. However, the interface of the groove ED joint exhibited defects, including local pores in the central region and partial grooves unfilled by molten material. This phenomenon reflects the limitations of the single groove ED structure in guiding the material flow, which tends to cause uneven joining quality at the interface and affects the overall connection stability. The grid ED joint exhibited a more pronounced interfacial energy concentration effect. In the initial melting stage, the reduced crystallinity of the grid ED structure lowered its rigidity, promoting the generation of interfacial viscoelastic heat to enable rapid melting of the surface layer. In the mixing stage, the intersecting design of the grid structure provided multidirectional flow channels for the

molten material, allowing it to flow and diffuse uniformly under ultrasonic vibration and pressure, whereas ultrasonic shearing induced interfacial crystallization, resulting in a slight increase in interfacial crystallinity after welding [43,45]. Owing to uniform melting, sufficient mixing, and moderately optimized crystallization behavior, the interfacial joining quality and uniformity were significantly superior to those of the other two structures, with carbon fibers tightly embedded in the molten resin matrix to form dense interfacial joining regions free of obvious pores caused by thermal decomposition or insufficient material filling, and only a few unfilled regions observed at the edges resulting from the boundary effect of material diffusion.



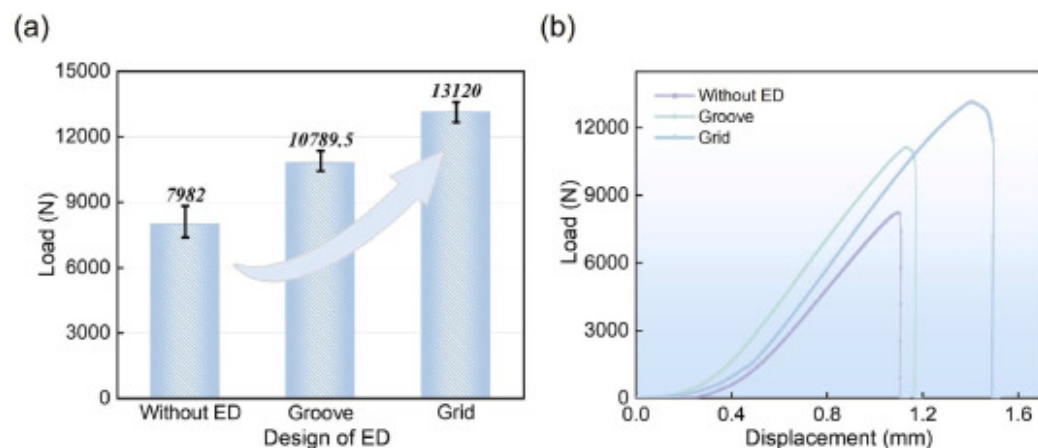
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Fig. 8. Microstructure of welded joints produced with different EDs: (a) without ED, (b) groove ED (1D), (c) grid ED (2D), (d) Micrograph of the region marked in (a), (e) Micrograph of the marked region in (b), (f) Micrographs of the marked region in (c).

3.3. Mechanical property and fracture morphology of different ED structure

A further analysis of the joint mechanical properties was conducted, and Fig. 9 systematically compares the mechanical performance differences among the without ED, linear groove ED (1D), and grid ED joints (2D). As shown in Fig. 9, the joint without ED exhibited scattered ultrasonic energy and insufficient interfacial heat generation during welding, resulting in a limited weld area formed by melting. The welds are mostly randomly distributed, making it difficult to form a continuous and dense joining region, resulting in poor overall interfacial joining. Consequently, the joint is prone to failure in these weak regions under stress, ultimately resulting in a low level of mechanical performance. In contrast, linear groove and grid EDs optimize the interfacial energy transfer and heat generation efficiency through structural design. The groove structure can guide the directional concentration of ultrasonic energy, reduce energy loss, significantly increase the effective joining area of the joint, effectively inhibit the formation of random weld spots, and promote the formation of continuous and uniform welds, thereby enhancing the overall joining strength of the joint. Under the same welding parameters, compared with the joint without ED, the failure load of the linear groove ED joint increased significantly by 35.2%, whereas the grid ED joint showed a more prominent improvement with an increase of 64.4%, reaching a peak failure load of 13,120N. This result verifies the superiority of the grid ED structure. Compared with the single linear groove, the cross design of the grid structure can more fully guide the energy concentration and material flow, form a more complete mechanical interlocking and interfacial joining, and ultimately exhibit a more excellent mechanical bearing capacity.

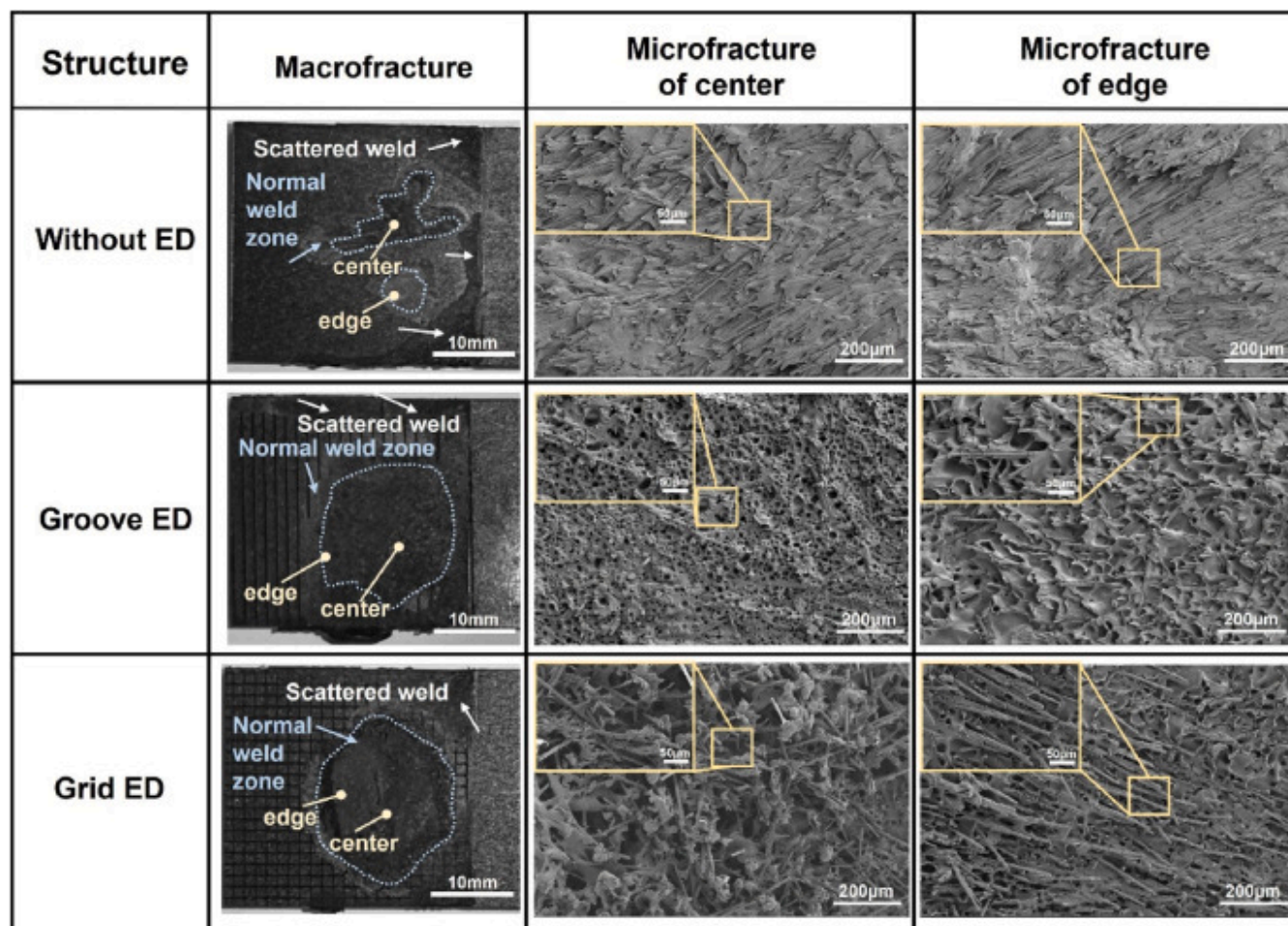


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Fig. 9. The mechanical property of joints (a) the joints failure load, (b) displacement load curve of the joints.

Fig. 10 shows the fracture morphologies of the joints without ED and joints with different ED structures. For the joint without ED, numerous random weld spots were observed on the fracture surface, with discontinuous weld spots in the central region. Further analysis of the microstructures at the edges and center revealed a morphology characterized by resin encapsulated carbon fibers owing to poor welding quality, indicating matrix dominated separation. Carbon fibers were embedded in the resin matrix without fiber pullout. In contrast, the fracture morphology of the linear grooved ED joints was significantly improved compared with that of the joints without ED, the number of randomly scattered weld spots on the fracture surface was significantly reduced, the resin melting area was expanded remarkably. However, it is worth noting that a large number of small pores are still distributed in the central welding zone, which is consistent with the previous observations of the interfacial microstructure. These dispersed microvoids tend to become stress concentration sources when the joint bears a shear load, leading to cracks initiating and propagating preferentially around the pores, thereby impairing the load bearing capacity and mechanical properties of the joint [46]. Compared with joints without ED and grooved ED joints, grid ED joints exhibited superior fracture characteristics. A large number of carbon fibers that were pulled out and exposed on the surface were observed at both the center and edge of the fracture region, confirming that the fracture mechanism of the joint is a mixed fracture mode dominated by fiber-matrix debonding [31]. Under shear loading, external forces first overcome the interfacial joining force between the fibers and resin matrix, leading to fiber matrix dejoining. The exposed carbon fibers enhance load transfer through interfacial friction and mechanical interlocking during stress application, retarding fracture propagation, which is the core reason for the excellent load bearing strength of the grid ED joint.



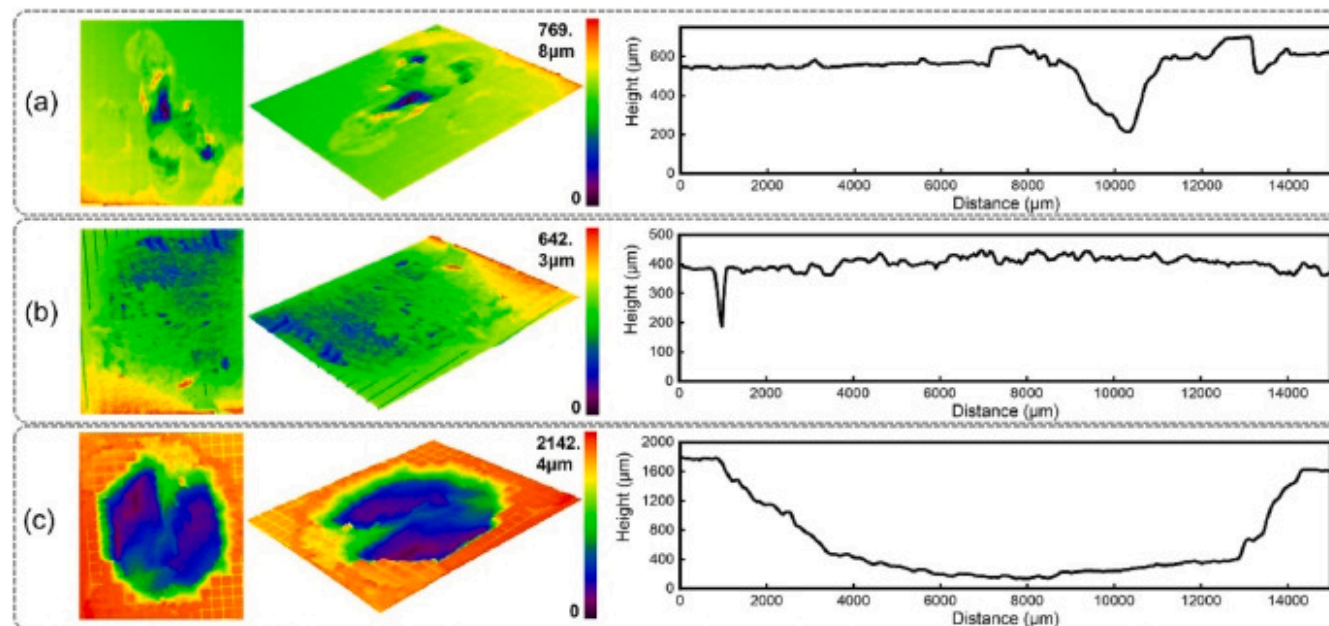
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Fig. 10. Fracture characteristics of joints with different EDs.

Fig. 11 shows the 3D morphologies and fracture paths of the joints without and with different ED structures. From the 3D morphological characterization results, the fracture surface of the joint without ED exhibited an obvious depression only in a small central weld spot area, whereas most other regions were relatively flat. The fracture surface of the groove ED joint was generally flat, whereas that of the grid ED joint exhibited a typical morphology of overall depression in the central weld spot

area, forming a distinct "basin-shaped" appearance. Further analysis of the fracture paths revealed that owing to the scattered interfacial energy and insufficiently concentrated heat generation during welding, the joint without the ED formed only a small area of effective weld spot in the central region, with most other areas being loosely joined interfacial transition zones. When the joint is subjected to shear loading, the small central weld spot struggles to bear the full external force and becomes a stress weak point. The fracture path initiated at the central weld spot and extended slowly toward the base material regions on both sides. In contrast, other regions, except the central weld spot, exhibited relatively flat fracture paths dominated by the interfacial fracture mode owing to the low interfacial joining strength, resulting in a poor overall load bearing capacity of the joint. For the groove ED joint, although the ED structure improves the energy concentration to a certain extent, weak joining defects such as pores still exist at the interface, leading to poor connection quality in the central weld spot area and a generally flat fracture path. In contrast, the grid ED joint exhibited distinctly different fracture behavior. Owing to the efficient energy concentration and material flow guidance of the grid structure, a large area, high quality continuous weld spot was formed in the central region of the joint. This region features tight interfacial joining and no obvious defects, making it resistant to direct damage from external forces. Thus, when the joint was loaded, the fracture path actively bypassed the stronger central high quality weld spot area and extended toward the base material regions with relatively lower mechanical properties on both sides. Ultimately, a cohesive fracture dominated by base material failure occurs, forming a distinctive "basin-shaped" fracture path. 3D morphological measurements indicate that the maximum depth of these depressed fracture paths reaches 0.18mm, and the depressed regions exhibit a dense and flat matrix structure, significantly enhancing the mechanical properties and fracture elongation of the joint.



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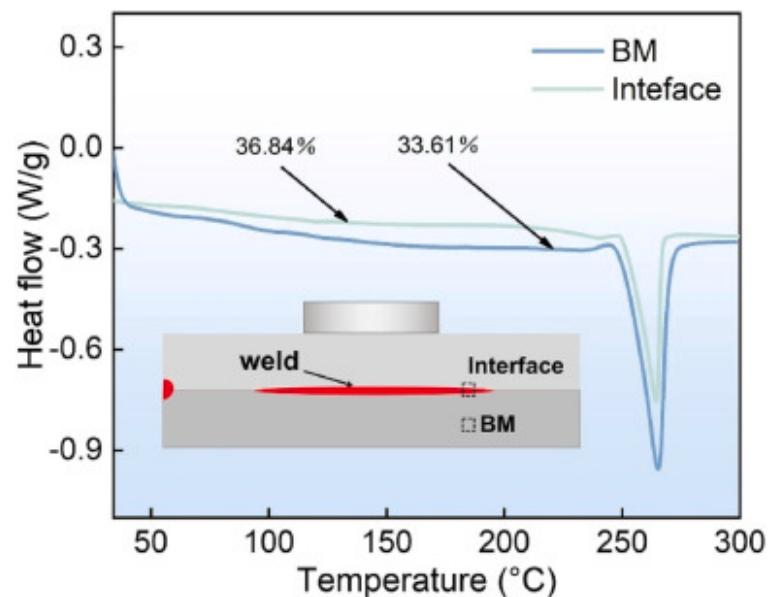
Fig. 11. 3D fracture morphology and fracture propagation paths of joints with different ED: (a) without ED, (b) groove ED (1D), (c) grid ED (2D).

3.4. Strengthening mechanism of laser etched ED

3.4.1. Crystallinity variation of manufactured EDs structure

As elaborated in Section 3.1, the reduced crystallinity of the laser-etched grid ED structure lowers its stiffness, thereby facilitating deformation and heat concentration during welding. Furthermore, the crystallinity at the interface changed after welding. Further analyze the inward buckling fracture characteristics of the grid ED structure, DSC was performed on the interfacial materials before and after welding, as shown in Fig. 12. Combined with the crystallinity analysis using Formula (3-1), the crystallinity of the post weld interfacial material was 36.84%, compared to 33.61% for the unwelded material, representing a 3.2% increase in the crystallinity of the melted interfacial material relative to the base material. This phenomenon is related to the

shear effect induced by high frequency vibration during ultrasonic welding and the slow cooling during the post weld pressure holding stage. The shear effect from high frequency vibration promotes the formation of regular crystal nuclei in the molten resin, whereas the post weld pressure holding stage allows the molten material to cool slowly within the crystallizable temperature range, providing sufficient time for the crystallization of the interfacial material [47]. This moderate increase in the crystallinity of the interfacial material promotes the formation of a denser interfacial structure and enhances the mechanical interlocking effect between the fibers and matrix [43,48]. The large area dense crystalline structure improves the strength of the interfacial material, making it prone to crack propagation through the matrix. The fracture surface adjacent to the base material exhibited layered resin tearing with clean fiber matrix separation, forming the aforementioned inward buckling "basin-shaped" fracture. The relatively straight and low roughness fracture path ultimately enhanced the mechanical properties of the joint.



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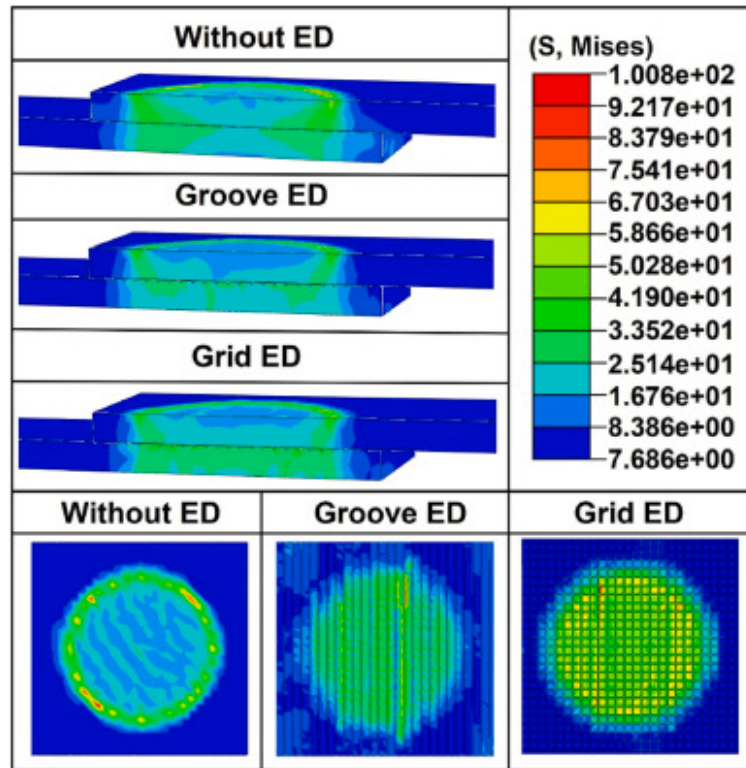
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Fig. 12. DSC curves of the interface of grid ED reinforcement joints and the base material.

3.4.2. Concentrated thermogenesis enhancement induced by ED

To further elucidate the thermomechanical effects of energy directors during ultrasonic welding, the welding process was analyzed in detail.

The stress distribution characteristics of joints with different energy director structures during the welding downward pressure stage were analyzed, and the corresponding stress results at the cross-section and interface are shown in [Fig. 13](#). For joints without an ED, forming an effective stress concentration at the interface is difficult. As can be seen from the interfacial stress figure, the joint overall exhibits a non uniform distribution with significantly high stress at the edges and low stress in the central region. The cross sectional stress distribution further confirmed that the stress was mainly concentrated at the edge regions where the horn directly contacted the upper plate, resulting in obvious stress dispersion and inhomogeneous distribution. In contrast, after introducing ED structures on the lower plate surface, the stress distribution of the joint was significantly altered: the originally dispersed interfacial stress was effectively guided and concentrated in the ED regions on the lower plate surface, laying a foundation for the subsequent preferential deformation and viscoelastic heat generation of the ED structures. Notably, compared with the grooved ED structure, the grid ED structure exhibited better uniformity in interfacial stress distribution, which helped improve the uniformity of the temperature field at the welding interface.



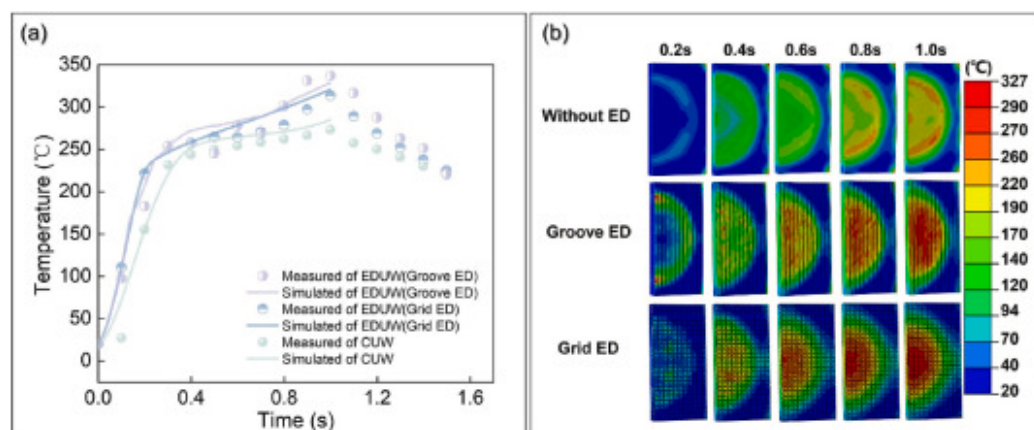
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Fig. 13. Stress distribution of different joints during horn downforce stage.

Fig. 14 illustrates the temperature evolution at the central interface of the joints under three conditions: groove EDs, grid EDs, and without EDs, during ultrasonic welding. It can be observed that the temperature curves obtained by numerical simulation are consistent with those measured experimentally in terms of the heating rate in the rising stage, the peak temperature range and the overall temperature characteristics, with only partial differences existing. These deviations mainly occur because the finite element model cannot fully and accurately simulate the microdefects on the surface and the actual contact conditions at the interface, along with the inherent lag and errors in thermocouple temperature measurement. This validates the applicability of the model for optimizing the process. Under the same welding parameters, in the initial stage of welding, the joint without ED fails to achieve effective energy concentration owing to the dispersion of ultrasonic energy at the interface, resulting in a

distinctly scattered temperature distribution. Heat was first generated and accumulated at the edge contact area between the horn and sheet, leading to a relatively slow overall heating rate. The central interface temperature gradually approached the material melting temperature (T_m) only after 0.5s into the welding process, accompanied by an uneven heat distribution. For the joint with groove EDs, heating was initiated at the horn-sheet contact interface. With increasing welding time, the energy concentrating effect of the groove structure caused a rapid increase in temperature at the protruding regions. However, the single directional groove structure leads to uneven heat distribution, which is prone to local thermal decomposition. In contrast, the joints with grid EDs exhibited distinctly different heating characteristics, benefiting from the uniform energy concentration and guidance enabled by the grid structure. Heat was rapidly concentrated in the central welding region during the initial stage of welding, eliminating the need for prolonged energy accumulation. The material melting temperature was reached within a short duration of 0.2–0.3s. More importantly, the joint temperature distribution was more uniform, preventing the thermal decomposition of the material. The temperature is maintained for an extended period within the range between the melting temperature (T_m) and the thermal decomposition temperature (T_d), facilitating the formation of high quality welded joints with tight bonding and minimal defect formation.

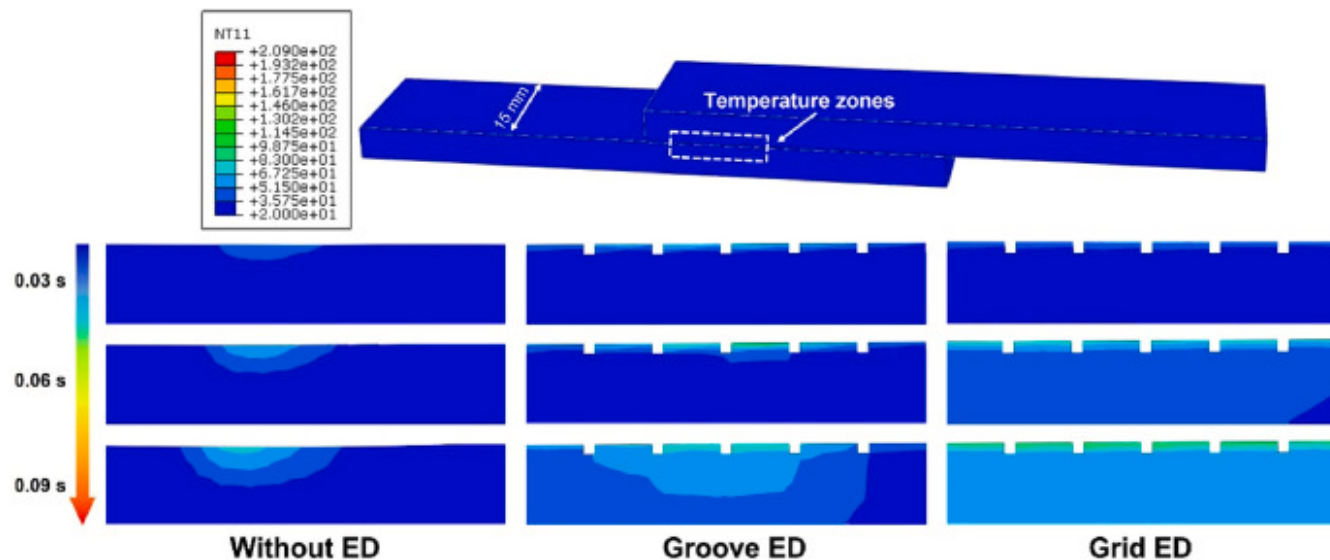


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Fig. 14. Ultrasonic welding processes of different welded joints: (a) Time-temperature curves (b) Simulated temperature evolution processes.

To further elucidate the heat generation mechanism of the ED protrusions at the edges of different joint cross-sections, the temperature distribution within the initial 0.1 s of ultrasonic welding was analyzed in detail, as illustrated in Fig. 15. In joints without an ED structure, the heat generated by ultrasonic vibration was initially confined to the edge regions in direct contact with the horn and subsequently propagated toward the center, resulting in a relatively limited heating zone owing to the flat interface and large contact area. Conversely, joints with groove and grid ED structures exhibited distinct thermal responses: heat rapidly accumulated at the protrusions, leading to significantly expanded heating zones and accelerated rates of temperature increase and propagation compared to the configuration without ED. This is primarily attributed to the drastically reduced actual contact area between the ED protrusions and upper plate, under the combined action of the high frequency vibration and axial pressure exerted by the horn, this localized reduction facilitates the preferential deformation of the ED regions, thereby expediting heat generation. Notably, compared with the groove ED structure, the grid ED structure demonstrated superior thermal uniformity by forming a more balanced temperature field rather than concentrating heat exclusively at the edges, a phenomenon also evident in Fig. 14.



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Fig. 15. Cross-sectional temperature distribution of different joints.

To elucidate the effect of the ED structure on interfacial welding energy, the frictional energy (Q_f) and viscoelastic energy (Q_v) during welding were analyzed. The heat generation from friction and viscoelasticity can be estimated using the following equations: (3-2) and (3-3) [[49], [50], [51]].

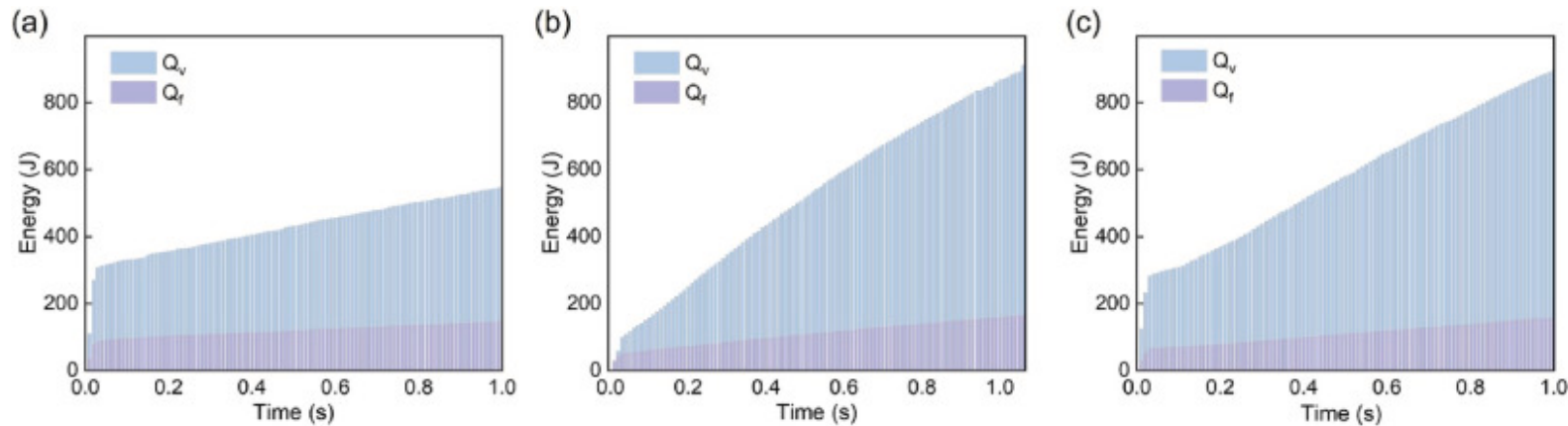
$$Q_f = e \frac{\omega}{\pi} \mu |\sigma_{yy}(\mathbf{x}) \mathbf{u}^*(\mathbf{x})| \quad (3-2)$$

where μ is the interfacial friction coefficient, $\sigma_{yy}(\mathbf{x})$ is the vertical contact stress, $\mathbf{u}^*(\mathbf{x})$ is the interfacial slip, and e is the hammer factor.

$$Q_v = e \frac{\omega \varepsilon^2 \mathbf{E}''}{2} \quad (3-3)$$

where $\mathbf{E}''$ denotes the loss modulus, ε the cyclic strain induced by mechanical vibration.

Fig. 16 illustrates the proportion of Q_f to Q_v during welding. In the initial welding stage, with the continuous action of ultrasonic vibration and the gradual enhancement of interfacial interaction, both frictional and viscoelastic energies exhibited a synchronous growth trend. However, their proportional relationship remained stable throughout, with the proportion of viscoelastic energy consistently higher than that of frictional energy. This result clearly indicates that viscoelastic heat generation dominates the heat generation mechanism during the ultrasonic welding of the CF/PA66 composites. Further comparison of the energy proportion among without ED, groove ED, and grid ED joints reveals that the incorporation of EDs exerts no significant influence on the magnitude or proportion of frictional energy. In striking contrast, the introduction of EDs markedly elevated both the total amount and proportion of viscoelastic energy. This phenomenon fully illustrates that the core function of ED structures at the welding interface lies in optimizing the interfacial energy transfer path and enhancing the strain concentration effect, thereby increasing the share of viscoelastic heat generation in the interfacial material and converting more ultrasonic energy into effective heat, which drives material melting. Moreover, a comparison of the energy values and proportions between the groove ED and grid ED joints indicates that the two configurations exhibit comparable energy magnitudes but differ in energy rise rates. The groove ED joint demonstrated a comparatively high energy rise rate, which readily resulted in a temperature maldistribution and localized overheating, as indicated in Fig. 14. In contrast, the grid ED joints feature a more moderate energy rise rate, which not only retains the energy concentration advantage inherent to EDs but also prevents localized overheating via the interconnectivity of the grid structure, thus achieving a more uniform energy distribution.



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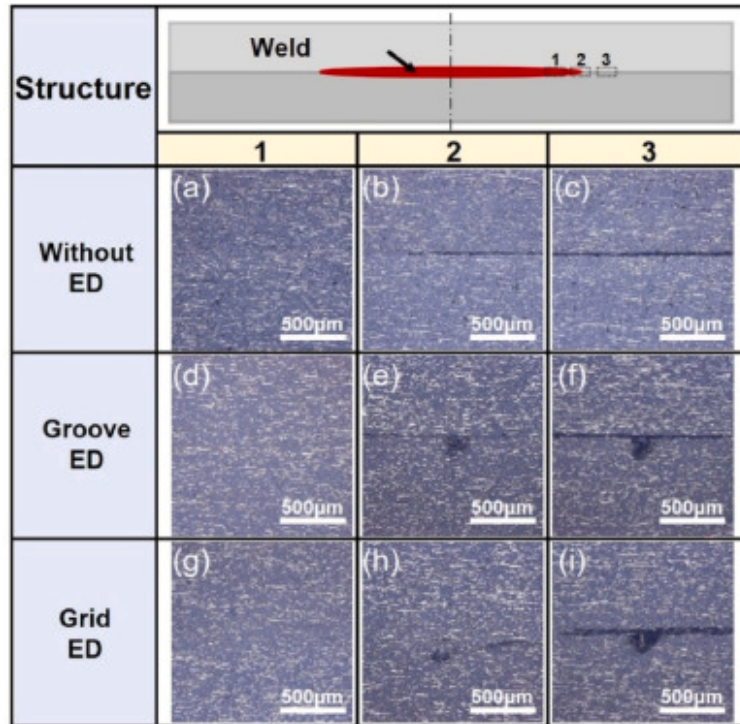
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Fig. 16. Simulated viscoelastic energy and friction energy: (a) without ED, (b) groove ED, (c) grid ED.

3.4.3. Strengthening by eliminating edge defects through the introduction of ED

Further experimental investigations serve to clarify the regulatory effects of EDs on the flow characteristics of molten materials. As illustrated in Fig. 17, a comparison of the edge material filling conditions among without ED joints, groove ED joints, and grid ED joints intuitively demonstrates the regulatory role of ED structures in molten material flow and interfacial filling. While joints with no ED, groove ED, and grid ED exhibited complete melting near the weld center, forming a tight, defect free interface with the upper plate, significant discrepancies emerged near the edge regions (regions 2 and 3). In the case of joints without an ED, the frictional and viscoelastic heat generated by the ultrasonic vibration, combined with the welding pressure, melted the resin. Lacking structural guidance, the molten material flowed primarily toward the edges along the lap interface, resulting in a flat, unfused defect at the perimeter (Fig. 17(b) and (c)). The introduction of the ED structure altered the flow path and filling behavior of the molten material. Driven by concentrated welding heat, localized temperatures increased rapidly to facilitates more complete melting of the matrix, thereby generating a tighter connection in region 2 (Fig. 17(e)–(h)), with the grid ED configuration proving superior to the groove structure. This advantage stems from the limited ability of unidirectional grooves to guide the flow of molten material, whereas the intersecting channels of the grid structure provide sufficient flow space for molten resin, promoting its uniform diffusion under welding pressure, this significantly improves the defect morphology at the

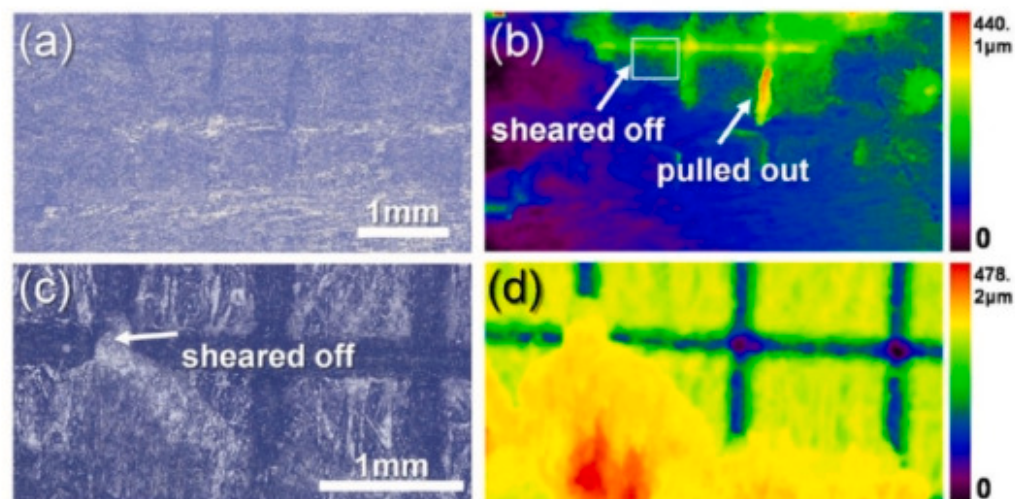
joint edges. When an external load is applied, unlike the throughout defects in without ED or grooved ED joints, crack propagation is restrained and extends into the matrix, thereby forming a cohesive fracture mode, as shown in Fig. 11. Meanwhile, the molten material filled in the edge grids undergoes shear fracture or pull out during failure, as illustrated in Fig. 18, which requires additional energy dissipation [7]. Such improvement in edge defects effectively enhances the overall mechanical performance of the grid ED joints.



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Fig. 17. Evolution of interface material filling in without ED, groove ED (1D), and grid ED joint (2D).



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Fig. 18. Fracture profile at the edge of the grid ED joint: (a) Groove filling material, (b) 3D morphology of (a), (c) Edge spreading material, (d) 3D morphology of (c).

4. Conclusion

This study innovatively adopts laser-etched surface treatment technology to fabricate groove typed (1D)/grid typed (2D) energy directors structures on CF/PA66 surfaces, achieving a significant performance enhancement. The effects of the ED structures on the joint microstructure, crystallinity, fracture morphology, and mechanical properties were systematically investigated, and the joint strengthening mechanism was elaborated in detail through finite element simulation analysis. The main conclusions are as follows.

- (1) In situ laser ablation was employed to form linear grooves and grid shaped energy director structures. Meanwhile, utilizing the rapid heating and cooling properties of the laser ablation process, the crystallinity of the material was altered as the two ED structures were fabricated at the interface. This reduced the rigidity in

geometrically abrupt regions and facilitated the generation of viscoelastic heat, thereby enhancing the energy utilization efficiency during welding.

- (2) ED structures fabricated via laser etching all improved interfacial fusion and significantly increased the joint failure load. However, the single groove ED tended to cause excessive local temperatures, generating micropores and thus deteriorating the joint mechanical properties. In contrast, the grid ED exhibited a superior improvement effect: it expanded the weld area while increasing crystallinity, demonstrated cohesive fracture characteristics with a "basin-shaped" fracture morphology, and achieved a joint failure load of 13,120N, representing a 64.4% increase compared to the joint without ED.
- (3) A finite element numerical simulation was employed to elucidate the strengthening effect of the ED structures on the joints from the perspective of stress and temperature evolution. Compared with the flat interface without EDs, the contact area between the ED protrusions and the upper plate was significantly reduced, leading to a local stress concentration, which in turn accelerated the generation of viscoelastic and frictional heat, enabling efficient heat accumulation in the key areas of the welding interface. In addition, the grid shaped ED structure has more advantages than the grooved ED structure, and its uniformly distributed grid morphology effectively avoids the problem of uneven interfacial temperature distribution caused by the linear groove ED structure.
- (4) Furthermore, for the grooved and grid structures fabricated on the plate surface by the in-situ laser ablation process, the recesses formed thereby not only facilitate energy concentration but also exert a guiding effect on the flow and diffusion of molten materials during the welding. Driven by the welding pressure, the molten resin material flows and spreads along the ablated recesses, which not only effectively promotes the fusion of molten materials at the interface and improves the continuity and compactness of the weld seam but also forms a mechanical interlocking structure at the bonding edge of the joint, thus further enhancing the joining strength.

CRedit authorship contribution statement

Jing Dong: Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Yuchun Liu:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Zulai Huang:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Zhiyu Xue:** Visualization, Validation, Methodology, Investigation. **Fenglei Sun:** Visualization, Validation, Methodology, Investigation. **Pengyu Zhong:** Visualization, Validation, Methodology, Investigation. **Li Zhou:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Yu Feng:** Resources, Project administration, Funding acquisition. **Baoming Wang:** Project administration, Funding acquisition. **Xiaoguo Song:** Writing – original draft, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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